

Evolution Analysis of Residual Stress in Cu/SiO₂ Hybrid Bonding During Bonding Process

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Abstract—Interface damage in Cu/SiO₂ hybrid bonding has emerged as a critical issue affecting its reliability, with residual stress induced during the manufacturing process being the primary contributor to such damage. Consequently, the investigation of interface residual stress has gained significant importance. In this study, a finite element simulation approach is employed to construct a Cu/SiO₂ hybrid bonding structure featuring an initial dishing depth. The residual stress at the interface, resulting from various manufacturing processes, is systematically analyzed. The results demonstrate that: (1) the residual stress at both the SiO₂/SiO₂ and Cu/Cu bonding interfaces is influenced by annealing temperature and the initial dishing depth of Cu; (2) after annealing, the residual stress at the SiO₂/SiO₂ bonding interface gradually increases as the interface approaches the Cu pad; and (3) the residual stress distribution at the Cu/Cu bonding interface after annealing exhibits central symmetry, with the maximum residual stress concentrated in the central region.

Keywords—Cu/SiO₂ hybrid bonding; numerical simulation; process flow; residual stress; reliability

I. INTRODUCTION

With the rapid development of semiconductor technology, the functionality of integrated circuits has become increasingly diversified. Applications such as remote work, artificial intelligence (AI), and automotive electronics are driving integrated circuits toward higher performance, higher integration, and greater reliability. As Moore's Law approaches its physical limits, front-end chip manufacturing has gradually reached a technological bottleneck, making it difficult to increase the number of transistors per unit area while maintaining cost efficiency. Consequently, there has been a growing international focus on advanced packaging technologies, aiming to overcome the limitations of Moore's Law through high-density interconnects in the back-end processes [1].

Among emerging packaging technologies, Cu/SiO₂ hybrid bonding is a highly promising technique. Cu/SiO₂ hybrid bonding eliminates the need for metal leads or microbumps, and instead achieves direct bonding through the interfaces between Cu/Cu and SiO₂/SiO₂. The bonding interconnects between SiO₂ layers provide mechanical support and electrical isolation for the entire integrated unit, while vertical electrical interconnections within the chip are achieved through the bonding of metal Cu layers. Due to its potential in

enabling ultra-high-density electrical interconnections (with pitch less than 10 μm), Cu/SiO₂ hybrid bonding is considered an ideal bonding solution for high-bandwidth memory (HBM) applications [2].

Tsau Y W [3] developed a specific structure for measuring Cu expansion and conducted experimental studies to investigate the expansion of Cu after bonding under different processing conditions. Gondcharton P [4] experimentally studied the impact of different bonding temperatures and Cu surface roughness on the initial bonding interface damage. To gain a deeper understanding of the mechanical behavior of the interface during the bonding process and optimize the process conditions, numerous studies have been conducted on Cu/SiO₂ hybrid bonding using advanced numerical simulation techniques.

Wang H [5] conducted finite element simulations of the Cu/Cu thermocompression bonding process, analyzing the effects of different bonding temperatures and various dishing depth values on the bonding area and interface. Beilliard Y [6] characterized Cu/Cu bonding by combining a cohesive interaction model with an elastoplastic model, studying its influence on the Cu/Cu interface bonding. Tippabhotla S K [7] established a numerical simulation model for chip-to-wafer hybrid bonding, investigating the bonding area and peeling stress of the bonding interface for Cu pads under both protruding and recessed conditions. Wlanis T [8] focused on the finite element modeling of copper surface roughness, combining the viscoplasticity and creep behavior of copper material to study the impact of Cu surface roughness on bonding outcomes. A series of studies by Ji L [9, 10, 11] established thermodynamic finite element models for different structures in Cu/SiO₂ hybrid bonding processes, considering various design parameters and process conditions such as dishing depth values, annealing temperature and time, and TSV (through-silicon via) pitch and depth. These studies provided two key results influencing bonding integrity: the Cu bonding area and interface peeling stress. However, these studies did not consider the stress effects caused by material deposition under high temperature and temperature variation conditions, which limits the ability to analyze the evolution of deformation from the perspective of the fundamental bonding process, thereby lacking a basis for process optimization.

During the Cu/SiO₂ hybrid bonding process, the differences in the materials' thermal expansion coefficients [12]

and the residual stresses induced by the manufacturing process are fundamental causes of interface damage. However, there is limited reporting on the influence of the fabrication process on the distribution of interface residual stresses. In this study, a finite element simulation method based on "dead and alive elements" is used to investigate the evolution of residual stresses during the Cu/SiO₂ hybrid bonding process. The simulation results presented in this paper aim to provide a foundation for the study of the interface damage mechanism.

II. PROCESS SIMULATION

A. Geometric Model

In this study, a three-dimensional finite element model was developed using ANSYS APDL software to simulate the bonding process of Cu/SiO₂ hybrid bonding structures. The Cu/SiO₂ hybrid bonding structure consists of three parts: a Si substrate, a SiO₂ insulating layer, and a Cu pad. In the Cu/SiO₂ hybrid bonding process, there is an initial dishing depth between Cu and SiO₂, with initial dishing depth values of 5 nm, 10 nm, and 12 nm for Cu. The three-dimensional model is shown in Figure 1(a), and an enlarged view of the dishing depth region is displayed in Figure 1(b). The dimensions of the Si substrate are 30×5×2 μm³, the dimensions of the left and right SiO₂ insulating layers are 13×1×2 μm³, and the Cu pad sizes are 4×0.995×2 μm³, 4×0.99×2 μm³, and 4×0.988×2 μm³.

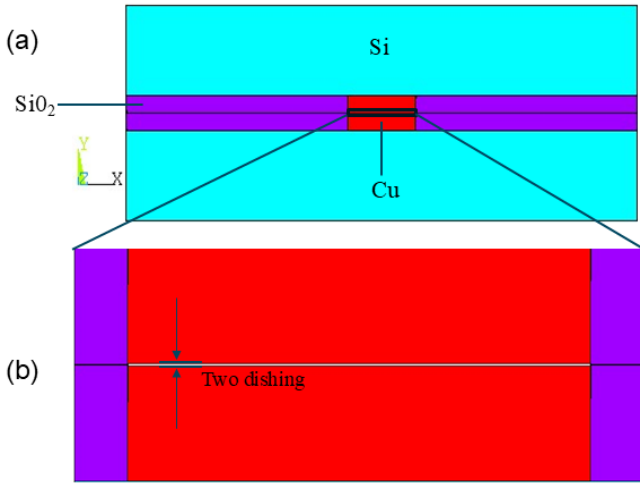


Fig. 1. (a) Schematic of Cu/SiO₂ hybrid bonding interposer packaging, (b) The amplify image of the dishing area.

B. Interface Behavior Assumptions

The results of this study focus on the stress distribution at the interface of the Cu/SiO₂ hybrid bonding structure under thermal loading. To achieve this, several assumptions were made during the simulation process:

- The Cu/Cu contact surface is assumed to be smooth, thereby eliminating the influence of surface roughness.
- The initial dishing depth of Cu is simplified as a flat concave surface, with the assumption that no cracking or separation occurs during the annealing process.
- To evaluate the effect of Cu expansion during annealing, it is assumed that the SiO₂-SiO₂ bonding does not experience any cracking during the annealing process.
- The Von Mises stress at the bonding interface is extracted during the bonding process to analyze the

impact of the bonding process on the residual stress distribution at the Cu/Cu and SiO₂-SiO₂ bonding interfaces.

C. Finite Element Model and Process Simulation Method

The three-dimensional finite element model established in this study is shown in Figure 2. It uses the SOLID185 solid element with 8 integration points, comprising 30,912 elements and 37,328 nodes. The reference temperature is set to 25°C. Three contact pairs are created using 3D 8-node surface-to-surface contact element (CONTA174) and 3D target element (TARGE170), which include the SiO₂ insulating layer on the left side, the SiO₂ insulating layer on the right side, and the Cu pad in the middle region. Considering the characteristics of the hybrid bonding array layout, the model is treated as one of the basic repeating units in the horizontal direction, with symmetry boundary conditions applied at the left and right ends. The top of the model is fixed in the z-direction, while the bottom is fixed in all three directions (x, y, z). This study focuses on the stress distribution at the Cu/Cu and SiO₂-SiO₂ interfaces and does not consider the stress distribution at the interface between the Si substrate and the Cu and SiO₂ layers.

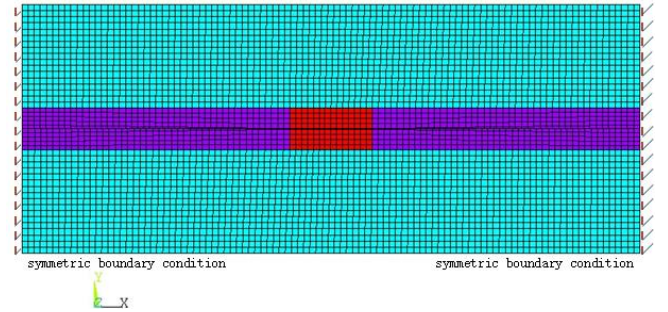


Fig. 2. FEM model of Cu/SiO₂ hybrid bonding package

The material parameters are presented in Table 1. The Si substrate and SiO₂ insulating layer are modeled using a linear elastic material model, while the Cu pad is modeled using an elastoplastic material model. During the annealing process, the Cu undergoes a high-temperature deformation phase where diffusion creep occurs at a sufficiently rapid rate. To simulate a more realistic process, the creep behavior of Cu is also taken into account in this simulation. The reference temperature is considered to be the deposition temperature of the material in a stress-free state. The yield stress of Cu is calculated using the Hall-Petch equation:

$$\sigma_y = \sigma_0 + \frac{k_y}{\sqrt{d}}$$

Where: d is the grain diameter of the polycrystalline material, σ_0 is initial stress value required to activate dislocation slip in polycrystalline materials, k_y is Hardening coefficient. For polycrystalline Cu^[13], the parameters σ_0 and k_y were set as 20 MPa and 0.14 MPa·m^{1/2}, respectively. The grain diameter d employed in this study was calculated from EBSD images acquired by Mirkarimi^[14], ultimately yielding a yield stress of 216.62 MPa for Cu.

The diffusion creep rate of Cu^[15] was calculated using the following formula:

$$\dot{\epsilon} = (K^D + K^S)^{-1} \sigma^A = \left(\frac{kT d^3}{10 V_{\text{atom}} w D_b} + \frac{9d}{10w} \eta \right)^{-1} \sigma^A$$

Table 1 Material properties

material	E(GPa)	ν	$\alpha(1/^\circ\text{C})$	Reference T($^\circ\text{C}$)	$\sigma_y(\text{MPa})$
Si	163	0.34	2.6e-6	25	
SiO ₂	73	0.17	0.55e-6	400	
Cu	128	0.34	1.65e-5	25	216.62

First, a finite element model incorporating the complete package structure was established. Subsequently, the elements corresponding to each component were activated or deactivated under the respective process load steps to simulate the component preparation process. The activation and deactivation of elements refer to the addition or removal of materials in the model, causing the corresponding elements to "live" or "die." Specifically, element activation is achieved by restoring the original values of element stiffness, mass, and element loads in the current load step. Element deactivation is accomplished by multiplying the element stiffness matrix by a minimal factor, at which point the loads, mass, stiffness, specific heat, and other similar properties of the deactivated elements are set to zero [16]. The process steps were simulated by applying the process temperature as a load across the entire finite element model. The temperature load time history for the assembly process is illustrated in Figure 3, and the process flow is depicted in Figure 4.

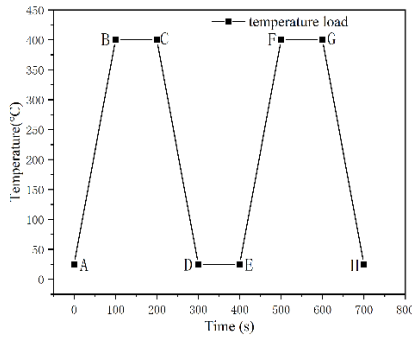


Fig. 3. Time history of temperature load

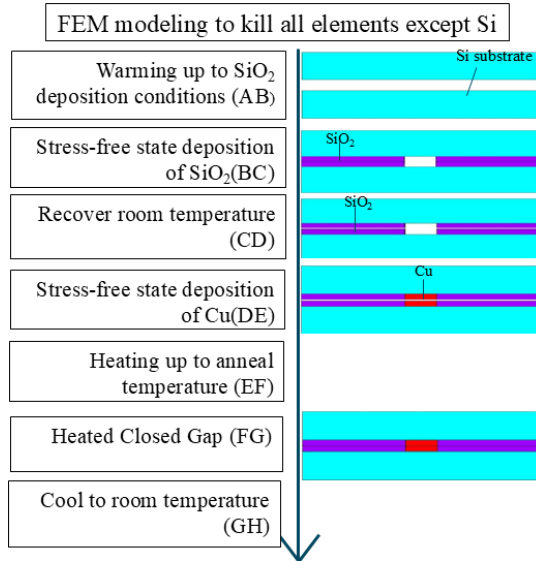


Fig. 4. The simulation steps of Process

Figure 4 illustrates the simulation steps, namely the Cu/SiO₂ hybrid bonding process flow:

(1) A finite element model containing the complete Cu/SiO₂ hybrid bonding structure was established, and all elements except the Si substrate were deactivated.

(2) The temperature was increased to meet the conditions required for SiO₂ deposition: the temperature was raised from room temperature (25 $^\circ\text{C}$) to 400 $^\circ\text{C}$ (as shown in the AB stage in Figure 3). The stress-free deposition temperature for SiO₂ is 400 $^\circ\text{C}$.

(3) SiO₂ was deposited under stress-free conditions (temperature load as shown in the BC stage in Figure 3): all SiO₂ elements in the relevant regions were activated to complete the SiO₂ deposition.

(4) The temperature was restored to room temperature: the Si substrate and the activated SiO₂ insulating layer were cooled from 400 $^\circ\text{C}$ to room temperature (25 $^\circ\text{C}$, as shown in the CD stage in Figure 3). The stress-free deposition temperature for Cu is 25 $^\circ\text{C}$.

(5) Cu was deposited under stress-free conditions (temperature load as shown in the DE stage in Figure 3): the elements in the intermediate region were activated to complete the Cu deposition.

(6) The temperature was increased to meet the heating conditions required for the annealing process: the temperature was raised from room temperature (25 $^\circ\text{C}$) to 400 $^\circ\text{C}$ (as shown in the EF stage in Figure 3). The annealing process temperature was set to 400 $^\circ\text{C}$.

(7) Gap closure through heating (as shown in the FG stage in Figure 3): a temperature load of 400 $^\circ\text{C}$ was applied to anneal the Cu/SiO₂ structure, closing the gaps to achieve bonding.

(8) Cooling to room temperature (25 $^\circ\text{C}$, as shown in the GH stage in Figure 3) completed the fabrication of the Cu/SiO₂ hybrid bonding structure.

III. RESULTS AND DISCUSSION

A. Mechanical Response of Hybrid Bonding

Figure 5 illustrates the evolution of residual mechanical responses during the bonding process under the conditions of a 10 nm initial recess and a 400 $^\circ\text{C}$ annealing temperature, the unit is MPa.

As the temperature increased from 25 $^\circ\text{C}$ to 400 $^\circ\text{C}$, as shown in Figures 5(a) to 5(b), the stress in the Si substrate rose from 0.76 MPa to 279.898 MPa. After the stress-free deposition of SiO₂ at 400 $^\circ\text{C}$, as depicted in Figure 5(c), the stress distribution did not exhibit significant changes. As the temperature decreased from 400 $^\circ\text{C}$ to 25 $^\circ\text{C}$, the stress distribution on the bonding surface of SiO₂ changed, with the maximum stress occurring at the corner where the SiO₂ surface connects to the Si substrate, as shown in Figure 5(d). Under the 25 $^\circ\text{C}$ condition, Cu deposition was performed, and the stress distribution remained largely unchanged, as illustrated in Figure 5(e). As the temperature increased from 25 $^\circ\text{C}$ to 400 $^\circ\text{C}$, noticeable deformation occurred on the surfaces of Cu and SiO₂, leading to the expansion and closure of gaps to achieve bonding, as shown in Figure 5(f)

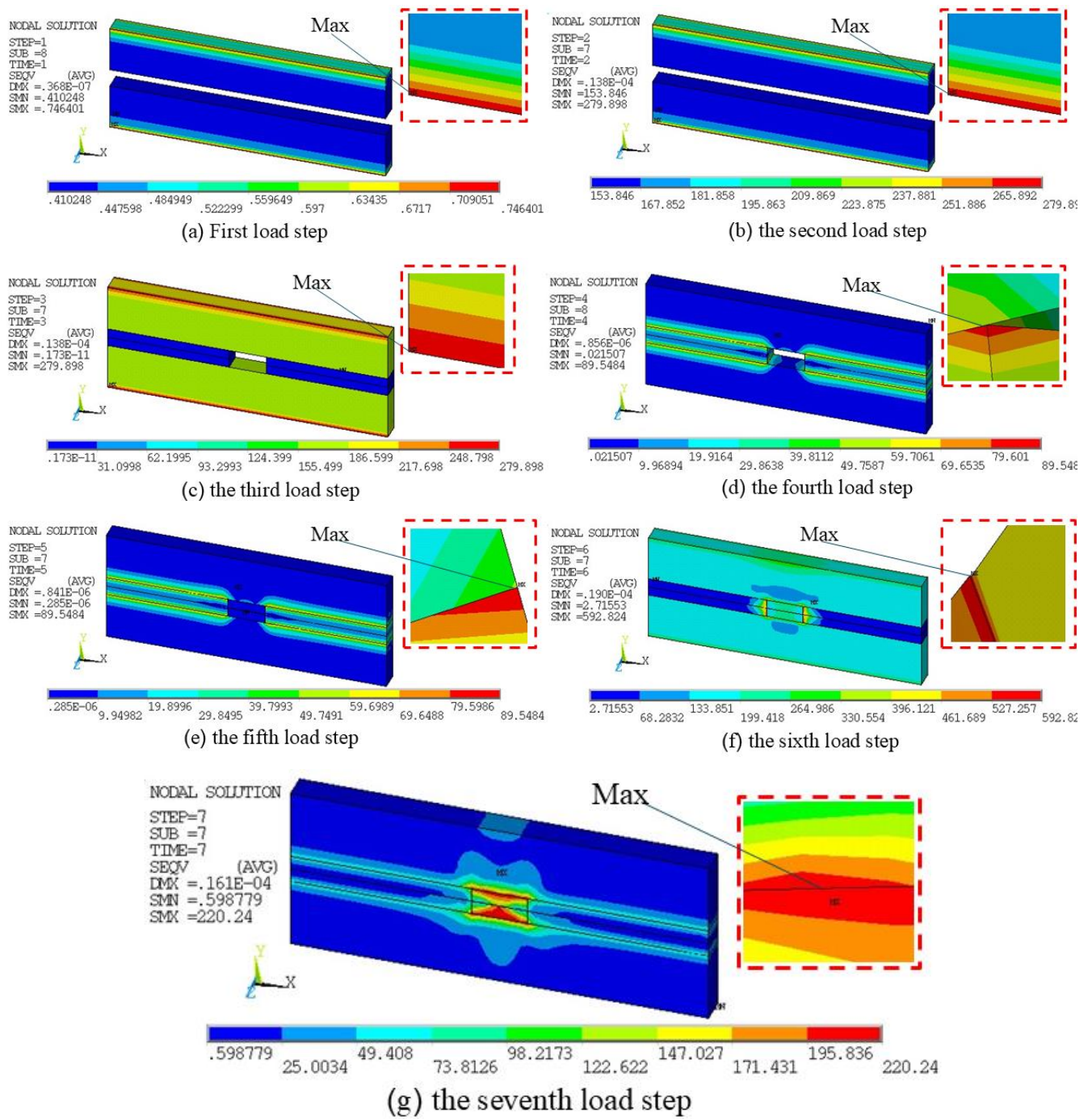


Fig. 5. mechanical response diagram

Finally, as the temperature cooled from 400°C to room temperature (25°C), the stress within the entire structure significantly decreased, with the maximum residual stress appearing at the connection between the upper Cu pad and the Si substrate, as depicted in Figure 5(g).

The bonding rate at the Cu interface is influenced by both temperature and the initial Cu dishing depth. Additionally, the residual stresses at the SiO₂ bonding interface and the Cu bonding interface after annealing are also affected by these factors. Although a reduction in the initial dishing depth increases the bonding area at the Cu interface, it simultaneously leads to an increase in residual stress at the SiO₂ interface. The maximum residual stress at the SiO₂ bonding interface occurs near the region adjacent to the Cu pad, while the maximum residual stress at the Cu

bonding interface is observed in the central area of the Cu pad.

B. Influence of Annealing Temperature and Dishing on Bonding Rate

The deformation diagram of the Cu/SiO₂ hybrid bonding structure after annealing is shown in Figure 6. It can be observed that the Cu/Cu interface is divided into bonded and non-bonded regions. The bonding rate can be determined by calculating the ratio of the bonded area to the diameter of the Cu pad.

l is defined as a line perpendicular to the bonding interface, and δ represents the difference in the y-coordinates between the node where line l intersects the lower surface of the upper Cu pad and the node where it

intersects the upper surface of the lower Cu pad. If $|\delta| > 0$, it indicates a non-bonded region, whereas if $|\delta| = 0$, it indicates a bonded region. Assuming the diameter of the Cu pad is d and the length of the bonded region is l_1 , the bonding rate A can be expressed as:

$$A = \frac{l_1}{d}$$

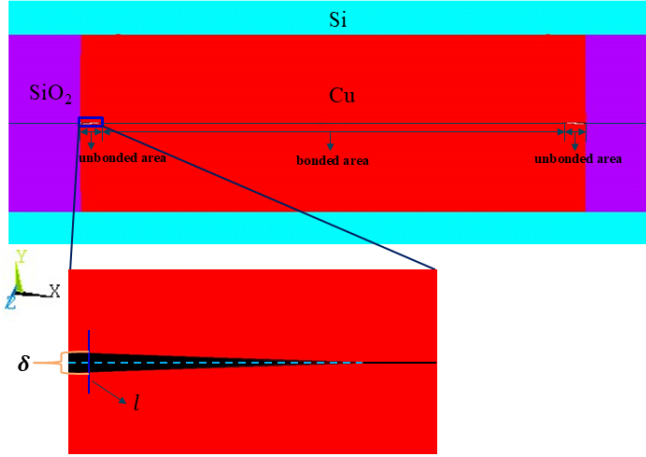


Fig. 6. Diagram of the deformation result

The variation of the bonding rate with temperature and initial dishing depth is shown in Figure 7. From the figure, it can be observed that the bonding rate increases with rising temperature. This is attributed to the enhanced thermal expansion of Cu at higher temperatures, which leads to an increase in the bonding area. Additionally, the bonding rate increases as the dishing depth decreases. This trend can be explained by the initial volume of Cu and its expansion behavior. As the dishing depth decreases, the available space for Cu to expand in the direction perpendicular to the contact surface is reduced. Consequently, Cu tends to expand more in the horizontal direction of the contact surface, resulting in a larger bonding area.

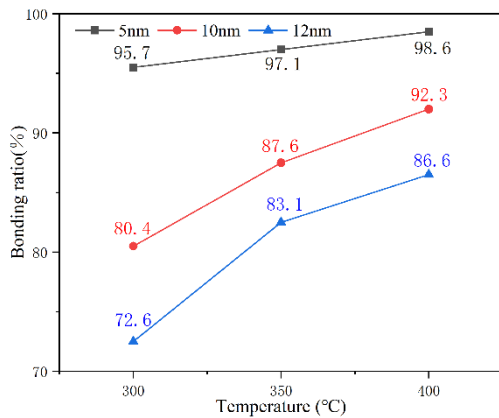


Fig. 7. Effect of annealing temperature and dishing on bonding ratio

C. Influence of Annealing Temperature and Dishing depth on Residual Stress

The residual stress at the left SiO₂/SiO₂ bonding interface as a function of the initial dishing depth and temperature is illustrated in Figure 8, while the residual

stress at the right SiO₂/SiO₂ bonding interface under the same conditions is shown in Figure 9.

The results indicate that under an annealing temperature of 400°C and an initial dishing depth of 5 nm, the peak residual stress at the SiO₂/SiO₂ bonding interface reaches its maximum value of 90.51 MPa. After annealing, the residual stress at the SiO₂/SiO₂ bonding interface gradually increases as it approaches the Cu pad, peaking at the region adjacent to the Cu pad. Furthermore, the residual stress increases with higher annealing temperatures. A smaller initial Cu dishing depth leads to greater residual stress at the SiO₂ bonding interface. Compared to the cases with initial dishing depths of 10 nm and 12 nm, the stress peak at the SiO₂ bonding interface adjacent to the Cu pad is the highest under the 5 nm initial recess condition. These stresses introduce additional damage at the SiO₂/SiO₂ interface, serving as the primary cause of integrity failure in the SiO₂/SiO₂ region near the Cu pad. This trend is consistent with the experimental observations by Ji H M^[17], who reported the initiation of voids at the SiO₂ interface near the Cu pad.

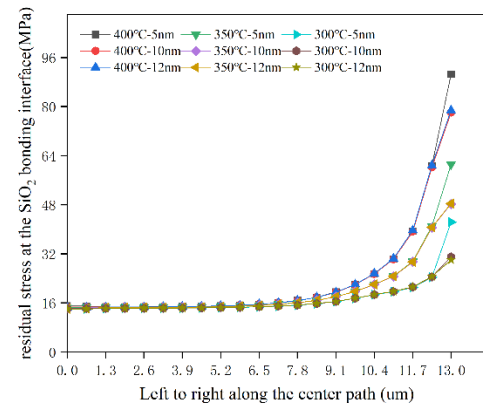


Fig. 8. Comparison of residual stresses at the SiO₂ interface on the left

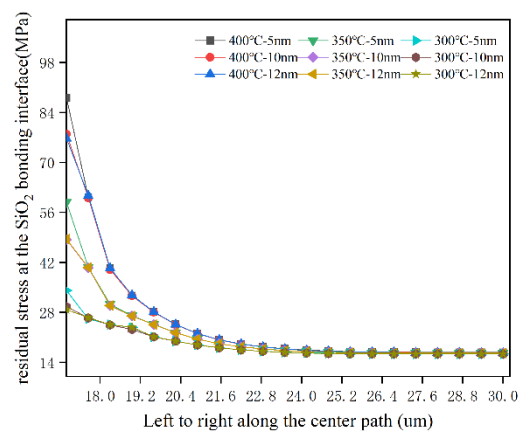


Fig. 9. Comparison of residual stresses at the SiO₂ interface on the right

The residual stress at the Cu/Cu bonding interface as a function of annealing temperature and initial dishing depth is shown in Figure 10.

The results indicate that the residual stress at the Cu/Cu bonding interface increases as the initial dishing depth decreases, and this effect is more pronounced with higher annealing temperatures. Specifically, the residual stress at the Cu/Cu bonding interface gradually rises with increasing annealing temperature. The residual stress distribution at the Cu/Cu bonding interface exhibits central symmetry, with the maximum residual stress occurring at the center of the bonded region on the Cu pad. This central area is the most likely location for initial interface failure, which is consistent with the experimental observations by Gondcharton P^[4], who reported the formation of voids at the center of the bonding interface.

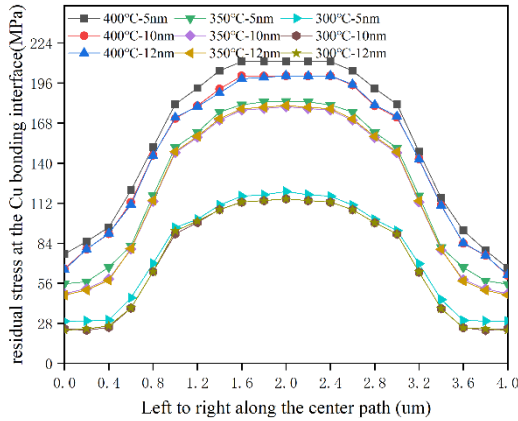


Fig. 10. Comparison of residual stresses at the Cu bonding interface

IV. CONCLUSION

In this study, a three-dimensional finite element analysis (FEA) modeling approach was employed to simulate the fabrication process of the Cu/SiO₂ hybrid bonding structure. The effects of the bonding process on the residual stress distribution at the SiO₂/SiO₂ and Cu/Cu interfaces were investigated. The main conclusions are as follows:

- 1) The bonding rate at the Cu/Cu interface is influenced by the annealing temperature and the initial Cu dishing depth. Higher annealing temperatures result in a higher bonding rate, while a smaller initial dishing depth also enhances the bonding rate.
- 2) After annealing, the residual stress at the SiO₂/SiO₂ bonding interface gradually increases as the interface approaches the Cu pad.
- 3) The residual stress distribution at the Cu/Cu bonding interface after annealing exhibits central symmetry, with the maximum residual stress concentrated in the central region.
- 4) To achieve a high bonding rate while avoiding excessive residual stress at the SiO₂/SiO₂ bonding interface, the recommended process parameters for sample preparation are an annealing temperature of 400°C and an initial dishing depth of 10 nm.

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